Hospital Patient Data Analysis

SQL Project | PostgreSQL | 10 Business Questions

1. Project Overview

This project analyses hospital records using SQL. A single table holds hospital, location, department, doctor and patient counts, admission and discharge dates and medical expenses. Ten business questions are answered using aggregate functions, GROUP BY, ORDER BY, LIMIT, date arithmetic and EXTRACT. The goal is to turn raw hospital data into insights on patient load, staffing, costs and length of stay.

2. Skills Demonstrated

Skill	Used for
CREATE TABLE / COPY	Designing the table and importing data from a CSV file
COUNT, SUM, AVG, MAX, ROUND	Totals, averages and maximum values
GROUP BY / ORDER BY / LIMIT	Grouping, ranking and Top-N results
Date arithmetic	Length of stay = Discharge_Date - Admission_Date
EXTRACT(month ...)	Monthly expense reporting

3. Table Structure

```
CREATE TABLE hos(
  Hospital_Name    VARCHAR(100),
  Location         VARCHAR(100),
  Department       VARCHAR(100),
  Doctors_Count    NUMERIC(10),
  Patients_Count   NUMERIC(10),
  Admission_Date   DATE,
  Discharge_Date   DATE,
  Medical_Expenses NUMERIC(10,2)
);
```

Data is loaded from a CSV file:

```
COPY hos(Hospital_Name, Location, Department, Doctors_Count, Patients_Count,
         Admission_Date, Discharge_Date, Medical_Expenses)
FROM 'Hospital_Data.csv' DELIMITER ',' CSV HEADER;
```

4. Business Questions and SQL Solutions

Q1. Total Number of Patients

Finds the total number of patients across all hospitals.

```
SELECT SUM(Patients_Count) AS total_patients
FROM hos;
```

Q2. Average Number of Doctors per Hospital

Shows the average doctor count for each hospital.

```
SELECT Hospital_Name,
       ROUND(AVG(Doctors_Count), 2) AS avg_doctors
FROM hos
GROUP BY Hospital_Name;
```

Q3. Top 3 Departments by Patients

Ranks departments by total patients and returns the top three.

```
SELECT Department,
       SUM(Patients_Count) AS total_patients
FROM hos
GROUP BY Department
ORDER BY total_patients DESC
LIMIT 3;
```

Q4. Hospital with Maximum Medical Expenses

Identifies the hospital with the highest recorded medical expenses.

```
SELECT Hospital_Name,
       MAX(Medical_Expenses) AS max_medical_expenses
FROM hos
GROUP BY Hospital_Name
ORDER BY max_medical_expenses DESC
LIMIT 1;
```

Q5. Daily Average Medical Expenses

Calculates the average expense per day of stay for each hospital.

```
SELECT Hospital_Name,
       ROUND(AVG(Medical_Expenses /
             NULLIF(Discharge_Date - Admission_Date, 0)), 2)
       AS avg_daily_expense
FROM hos
GROUP BY Hospital_Name
ORDER BY avg_daily_expense DESC;
```

Q6. Longest Hospital Stay

Finds the record with the longest stay using the date difference.

```
SELECT Hospital_Name, Department,
       Admission_Date, Discharge_Date,
       (Discharge_Date - Admission_Date) AS stay_days
FROM hos
ORDER BY stay_days DESC
LIMIT 1;
```

Q7. Total Patients Treated per City

Counts patients treated in each city.

```
SELECT Location,
       SUM(Patients_Count) AS total_patients
FROM hos
GROUP BY Location
ORDER BY total_patients DESC;
```

Q8. Average Length of Stay per Department

Calculates the average number of days spent in each department.

```
SELECT Department,
       ROUND(AVG(Discharge_Date - Admission_Date), 2)
       AS avg_stay_days
FROM hos
GROUP BY Department
ORDER BY avg_stay_days DESC;
```

Q9. Department with the Lowest Number of Patients

Finds the department with the fewest patients overall.

```
SELECT Department,
       SUM(Patients_Count) AS total_patients
FROM hos
GROUP BY Department
ORDER BY total_patients ASC
LIMIT 1;
```

Q10. Monthly Medical Expenses Report

Groups records by admission month and totals the expenses.

```
SELECT EXTRACT(MONTH FROM Admission_Date) AS month,
       SUM(Medical_Expenses) AS total_medical_expenses
FROM hos
GROUP BY EXTRACT(MONTH FROM Admission_Date)
ORDER BY month;
```

5. Key Learnings

- Choosing the right aggregate (SUM vs COUNT vs AVG) changes the meaning of the answer.
- GROUP BY must match the level at which the result is needed (hospital, department or city).
- Date subtraction in PostgreSQL returns days directly, which makes length-of-stay easy.
- NULLIF protects against divide-by-zero for same-day admissions.
- ORDER BY with LIMIT is the simple pattern for Top-N and Bottom-N questions.

Tools: PostgreSQL, SQL

Learned under the guidance of Sathish Dhawale Sir, Founder of Skill Course. Thank you for the practical, project-based teaching.